Lesson 3.11 - How to Calculate the Standard Deviation

> **Video:** How To Calculate The Standard Deviation by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formula (sample standard deviation)

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Steps

1. Find the **mean** $\bar{x}$.
2. For each value, find the **deviation** $x - \bar{x}$.
3. **Square** each deviation. The sign stops mattering, so you can use the distance from the mean.
4. **Add** the squared deviations.
5. **Divide** by $n - 1$.
6. Take the **square root**.

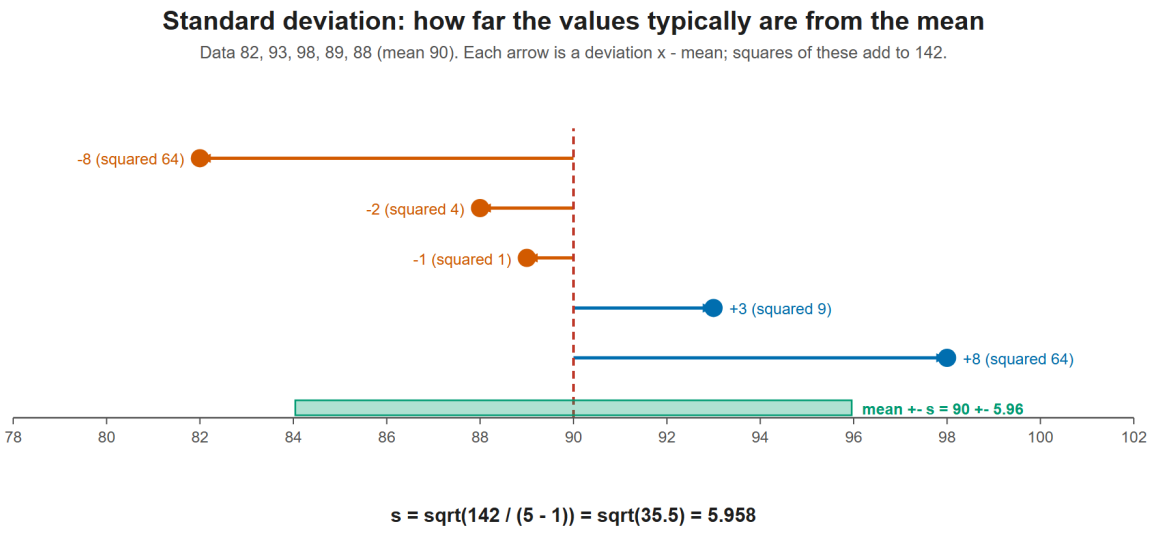
Example 1 - five values

Data: 82, 93, 98, 89, 88

Mean: $\bar{x} = \frac{\sum x}{n} = \frac{450}{5} = 90$

x	$x - \bar{x}$	$(x - \bar{x})^2$
82	−8	64
93	3	9
98	8	64
89	−1	1
88	−2	4
Sum	0	**142**

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}} = \sqrt{\frac{142}{5 - 1}} = \sqrt{35.5} \approx 5.958$$



Each arrow is one deviation from the mean of 90. Squaring them (64, 9, 64, 1, 4) and adding gives 142. Dividing by n - 1 = 4 and taking the square root gives s = 5.96. The green band shows mean +- s.

Example 2 - ten values

Data: 76, 84, 69, 92, 58, 89, 73, 97, 85, 77

Mean: $\bar{x} = \frac{\sum x}{n} = \frac{800}{10} = 80$

Shortcut: squaring removes the sign, so just write how far each value is from 80.

x	Distance from 80	Squared
76	4	16
84	4	16
69	11	121
92	12	144
58	22	484
89	9	81
73	7	49
97	17	289
85	5	25
77	3	9
Sum		**1234**

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}} = \sqrt{\frac{1234}{10 - 1}} = \sqrt{137.11} \approx 11.709$$

Key takeaways

- Standard deviation measures the **typical distance of the data from the mean**.
- The sample version divides by $n - 1$. The population version (σ) divides by N and uses μ.
- Check: the deviations $x - \bar{x}$ always add up to **0**.